PROPERNESS TESTED ON A^N-BASE CHANGES (LIMITED BASE CHANGE)

THE STACKS PROJECT AUTHORS

Editorial note. The env carries two proof blocks; the first is a pointer ('a special case of Lemma test-universally-closed', itself a delivered card) and is disclosed here; the second proof is the substantial one and both are included in the counted span.

Allowed standard prerequisites (editorial).

- Chow's lemma for finite type. Limits, Lemma lemma-chow-finite-type (same chapter)
- Image proper is proper. Morphisms, Lemma morphisms-lemma-image-proper-is-proper
- Standard covering of projective space. Constructions, Lemma constructions-lemma-standard-covering-projective-space

Source reference. Original lemma 05JX: sources/stacks-chow/tags.txt, source lines 7181–7181. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01W2: sources/stacks-chow/tags.txt, source lines 2467–2467. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0202: sources/stacks-chow/tags.txt, source lines 2619–2619. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01W3: sources/stacks-chow/tags.txt, source lines 2468–2468. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01W6: sources/stacks-chow/tags.txt, source lines 2471–2471. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01IQ: sources/stacks-chow/tags.txt, source lines 1987–1987. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03GN: sources/stacks-chow/tags.txt, source lines 4534–4534. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01NG: sources/stacks-chow/tags.txt, source lines 2157–2157. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01W4: sources/stacks-chow/tags.txt, source lines 2469–2469. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 01KT: sources/stacks-chow/tags.txt, source lines 2062–2062. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let $f : X \to S$ be a separated morphism of finite type. The following are equivalent:*

1. *The morphism f is proper.*
2. *For any morphism $S' \to S$ which is locally of finite type the base change $X_{S'} \to S'$ is closed.*
3. *For every $n \geq 0$ the morphism $\mathbf{A}^n \times X \to \mathbf{A}^n \times S$ is closed.*

First proof. In view of the fact that a proper morphism is the same thing as a separated, finite type, and universally closed morphism, this lemma is a special case of Lemma 05JX. □

Second proof. Clearly (1) implies (2), and (2) implies (3), so we just need to show (3) implies (1). First we reduce to the case when S is affine. Assume that (3) implies (1) when the base is affine. Now let $f : X \to S$ be a separated morphism of finite type. Being proper is local on the base (see Morphisms, Lemma 01W2), so if $S = \bigcup_\alpha S_\alpha$ is an open affine cover, and if we denote $X_\alpha := f^{-1}(S_\alpha)$, then it is enough to show that $f|_{X_\alpha} : X_\alpha \to S_\alpha$ is proper for all α. Since S_α is affine, if the map $f|_{X_\alpha}$ satisfies (3), then it will satisfy (1) by assumption, and will be proper. To finish the reduction to the case S is affine, we must show that if $f : X \to S$ is separated of finite type satisfying (3), then $f|_{X_\alpha} : X_\alpha \to S_\alpha$ is separated of finite type satisfying (3). Separatedness and finite type are clear. To see (3), notice that $\mathbf{A}^n \times X_\alpha$ is the open preimage of $\mathbf{A}^n \times S_\alpha$ under the map $1 \times f$. Fix a closed set $Z \subset \mathbf{A}^n \times X_\alpha$. Let $\bar{Z}$ denote the closure of Z in $\mathbf{A}^n \times X$. Then for topological reasons,

$$1 \times f(\bar{Z}) \cap \mathbf{A}^n \times S_\alpha = 1 \times f(Z).$$

Hence $1 \times f(Z)$ is closed, and we have reduced the proof of (3) $\Rightarrow$ (1) to the affine case.

Assume S affine, and $f : X \to S$ separated of finite type. We can apply Chow's Lemma 0202 to get $\pi : X' \to X$ proper surjective and $X' \to \mathbf{P}^n_S$ an immersion. If X is proper over S, then $X' \to S$ is proper (Morphisms, Lemma 01W3). Since $\mathbf{P}^n_S \to S$ is separated, we conclude that $X' \to \mathbf{P}^n_S$ is proper (Morphisms, Lemma 01W6) and hence a closed immersion (Schemes, Lemma 01IQ). Conversely, assume $X' \to \mathbf{P}^n_S$ is a closed immersion. Consider the diagram:

$$\begin{array}{ccc} X' & \longrightarrow & \mathbf{P}^n_S \\ {\scriptstyle\pi}\downarrow & & \downarrow \\ X & \xrightarrow{f} & S \end{array} \tag{0.1.1}$$

All maps are a priori proper except for $X \to S$. Hence we conclude that $X \to S$ is proper by Morphisms, Lemma 03GN. Therefore, we have shown that $X \to S$ is proper if and only if $X' \to \mathbf{P}^n_S$ is a closed immersion.

Assume S is affine and (3) holds, and let n, X', π be as above. Since being a closed morphism is local on the base, the map $X \times \mathbf{P}^n \to S \times \mathbf{P}^n$ is closed since by (3) $X \times \mathbf{A}^n \to S \times \mathbf{A}^n$ is closed and since projective space is covered by copies of affine n-space, see Constructions, Lemma 01NG. By Morphisms, Lemma 01W4 the morphism

$$X' \times_S \mathbf{P}^n_S \to X \times_S \mathbf{P}^n_S = X \times \mathbf{P}^n$$

is proper. Since $\mathbf{P}^n$ is separated, the projection

$$X' \times_S \mathbf{P}^n_S = \mathbf{P}^n_{X'} \to X'$$

will be separated as it is just a base change of a separated morphism. Therefore, the map $X' \to X' \times_S \mathbf{P}^n_S$ is proper, since it is a section to a separated map (see Schemes, Lemma 01KT). Composing these morphisms

$$X' \to X' \times_S \mathbf{P}^n_S \to X \times_S \mathbf{P}^n_S = X \times \mathbf{P}^n \to S \times \mathbf{P}^n = \mathbf{P}^n_S$$

we find that the immersion $X' \to \mathbf{P}^n_S$ is closed, and hence a closed immersion. □